

# KRITTI SHARMA

PhD Candidate in Astrophysics, California Institute of Technology

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## EDUCATION

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**California Institute of Technology**

Pasadena, CA, USA

*PhD in Astrophysics*

2023-Present

**Advisor:** Prof. Vikram Ravi

**Thesis:** “*Fast Radio Bursts: Cosmic Beacons Illuminating the Baryonic Matter in the Universe*”

**Research Interests:** I investigate the distribution of baryons to study the physics of galaxy formation, with the broader goal of constraining fundamental cosmological parameters. I combine complementary probes, including FRB dispersion measures, the kinematic Sunyaev-Zel’dovich effect, and galaxy populations, to decipher how baryons shape the underlying cosmic web and to mitigate baryonic uncertainties in next-generation cosmological analyses.

**California Institute of Technology**

Pasadena, CA, USA

*MS in Astrophysics*

2022-2023

**Selected Coursework:** Radiative Processes, Astrophysical Cosmology, Structure and Evolution of Galaxies, Interstellar and Intergalactic Medium, High-Energy Astrophysics, Structure and Evolution of Stars, Astronomical Measurements and Instrumentation, Fundamentals of Statistical Learning

**Indian Institute of Technology Bombay**

Mumbai, MH, India

*B.Tech in Mechanical Engineering with Honors, and Minor in Computer Science*

2018-2022

**Advisors:** Prof. Varun Bhalerao, Prof. Archana Pai

**Theses:** “*Convolutional Neural Network-based Algorithm for Detecting Intermediate-mass Black Hole Binaries in LIGO Data*” and “*Near-Earth Asteroid Observations and Data Reduction Software for GROWTH-India Telescope*”

**Selected Coursework:** Astrophysics, Special Theory of Relativity, Classical Mechanics, Quantum Physics, Electricity and Magnetism, Advanced Machine Learning, Intelligent & Learning Agents, Deep Learning, Programming for Data Science, Data Structures, Algorithms, Operating Systems, Differential Equations, Linear Algebra, Automatic Speech Recognition, Digital Image Processing

## OBSERVING EXPERIENCE

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**W.M. Keck Observatory, Maunakea on Hawai’i**

2022-Present

**27 nights** (Co-I; PI: Ravi, Connor, Ocker) on LRIS, MOSFIRE, KCWI, DEIMOS, and ESI to build a complete catalog of 40+ Fast Radio Burst (FRB) host galaxies localized by Deep Synoptic Array.

**Hale Telescope, Palomar Observatory**

2022-2025

**16 nights** (Co-I; PI: Ravi, Connor) on DBSP, NGPS, WaSP, and WIRC, to complement FRB host Keck observations.

**GROWTH-India Telescope, Indian Astronomical Observatory**

2020-2022

**50+ nights** to follow up 70+ Asteroids, 30+ Comets, and 4 Gamma Ray Bursts, as a part of GROWTH collaboration.

## RESEARCH PUBLICATIONS

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**Refereed publications:** 35 (11 first authored; 4 second/third authored; 20 co-authored)

**Non-refereed publications:** 210 (1 [AAS Research Note](#), 186 [MPECs](#), 16 [ATels](#), and 7 [GCNs](#) for discovery and follow-up observations of minor planets, comet outbursts, and Gamma Ray Burst afterglows)

**Citations (all papers):** 1,002

**Citations (first author):** 223

**H index:** 16

### First-Author Papers

11. “*Backlighting the Universe with Fast Radio Bursts: An Anthology of Dispersion Measure Cross Correlations with Tracers of Large-scale Structure and Baryonic Matter*”  
K. Sharma, E. Krause, V. Ravi, *et al.*, 2026, ApJ, 1010, 47. [arXiv:2604.22105](#).
10. “*Signatures of Suppressed Matter Clustering revealed by Fast Radio Bursts*”  
K. Sharma, E. Krause, V. Ravi, *et al.*, 2026, Nature Astronomy, 10, 2397-3366. [arXiv:2604.17162](#).

9. “*Baryons in the Darkest Sites of the Universe*”  
K. Sharma, V. Ravi, D. Anbajagane, *et al.*, 2026, ApJL, 1006, L3. [arXiv:2605.01994](#).
8. “*Quantifying the Impact of Selection Effects on FRB DM– $z$  Relation Cosmological Inference*”  
K. Sharma, V. Ravi, E. Krause, *et al.*, 2026, ApJ, 999, 202. [arXiv:2511.16850](#).
7. “*Probing Baryonic Feedback and Cosmology with the  $3 \times 2$ –point Statistic of FRBs and Galaxies*”  
K. Sharma, E. Krause, V. Ravi, *et al.*, 2026, ApJ, 998, 109. [arXiv:2509.05866](#).
6. “*A Hydrodynamical Simulations-based Model that Connects the FRB DM–Redshift Relation to Suppression of the Matter Power Spectrum via Feedback*”  
K. Sharma, E. Krause, V. Ravi, *et al.*, 2025, ApJ, 989, 81. [arXiv:2504.18745](#).
5. “*Fast Luminous Extragalactic Transients in the VLA Sky Survey: Implications for the Rates of Accretion-induced Collapse Events, Fast Blue Optical Transients and Gamma Ray Burst Afterglows*”  
K. Sharma, V. Ravi, D. Dong, *et al.*, 2025, PASP, 137, 084102. [arXiv:2506.04522](#).
4. “*Preferential Occurrence of Fast Radio Bursts in Massive Star-forming Galaxies*”  
K. Sharma, V. Ravi, L. Connor, *et al.*, 2024, Nature, 635, 61–66. [arXiv:2409.16964](#).
3. “*Searching for Massive Black Hole Binaries with a Transfer Learning Algorithm*”  
K. Sharma, K. Chandra, and A. Pai, 2023, Phys. Rev. D 108, 124061. [arXiv:2208.02545](#).
2. “*Astreaks: Astrometry of NEOs with Trailed Background Stars*”  
K. Sharma, H. Kumar, H. Choudhary, *et al.*, 2023, MNRAS, 524, 2, 2651–2660. [arXiv:2306.16519](#).
1. “*A Massive Elliptical Host Among Two Galaxy-cluster Fast Radio Bursts*”  
K. Sharma, J. Somalwar, C. Law, *et al.*, 2023, ApJ, 950, 175. [arXiv:2302.14782](#).

## Second/Third-Author Papers

4. “*A new measurement of the FRB DM–galaxy cross correlation and a first joint analysis with the kinematic SZ effect*”  
S. McCarty, L. Connor, K. Sharma, *et al.* 2026, under review in PRD. [arXiv:2608.11296](#).
3. “*A Search for Great Walls of Cosmic Baryons in the Northern Sky*”  
V. Ravi, K. Sharma, and L. Connor, 2026, ApJL, 1008, L57. [arXiv:2604.05093](#).
2. “*A gas-rich cosmic web revealed by the partitioning of the missing baryons*”  
L. Connor, V. Ravi, K. Sharma, *et al.*, 2025, Nature Astronomy, 9, 1226–1239. [arXiv:2409.16952](#).
1. “*Deep Synoptic Array Science: First FRB and Host Galaxy Catalog*”  
C. Law and K. Sharma, *et al.*, 2024, ApJ, 967, 29. [arXiv: 2307.03344](#).

## Papers with Supporting Contributions

20. “*Tracing the Host-Galaxy Environments of FRBs using Localized Sources Detected with the DSA*”  
F. Verdi, V. Ravi, *et al.* (incl. K. Sharma), under review in ApJ. [arXiv:2609.30616](#).
19. “*A multi-telescope fit to the FRB population, allowing for FRB repetition*”  
J. Hoffman, C. James, *et al.* (incl. K. Sharma), under review in PASA. [arXiv:2609.27282](#).
18. “*The DSA/Chronoscope fast radio burst survey: forecasts and science overview*”  
L. Connor, K. Shin, *et al.* (incl. K. Sharma), under review in OJAp. [arXiv:2608.30943](#).
17. “*Searching for Long Duration Radio Transients with the DSA-110*”  
M. Sherman, N. Kosogorov, *et al.* (incl. K. Sharma), 2026, PASP, 138, 024501. [arXiv:2510.18136](#).
16. “*A Luminous and Hot Infrared through X-Ray Transient at a 5 kpc Offset from a Dwarf Galaxy*”  
J. Somalwar, V. Ravi, *et al.* (incl. K. Sharma), 2025, ApJ, 995, 228. [arXiv:2505.11597](#).
15. “*Optical/Infrared Observations of the Extraordinary GRB 250702B: A Highly Obscured Afterglow in a Massive Galaxy Consistent with Multiple Possible Progenitors*”  
J. Carney, I. Andreoni, *et al.* (incl. K. Sharma), 2025, ApJL, 994, L46. [arXiv:2509.22784](#)
14. “*A Correlation between Fast Radio Burst Dispersion Measure and Foreground large-scale Structure*”  
M. Hussaini, L. Connor, *et al.* (incl. K. Sharma), 2025, ApJL, 993, L27. [arXiv:2506.04186](#)
13. “*Evidence for a Hot Galactic Halo around the Andromeda Galaxy Using FRBs along Two Sightlines*”  
R. Anna-Thomas, C. Law, *et al.* (incl. K. Sharma), 2025, ApJ, 993, 221. [arXiv:2503.02947](#).

12. [“A 50 Mpc Fast Radio Burst Constrains the Mass of the Milky Way Circumgalactic Medium”](#)  
V. Ravi, M. Catha, *et al.* (incl. K. Sharma), 2025, AJ, 169, 330, [arXiv:2301.01000](#).
11. [“The Palomar twilight survey of ‘Ayló’chaxnim, Atras, and comets”](#)  
B. Bolin, F. Masci, *et al.* (incl. K. Sharma), 2025, Icarus, 425, 116333. [arXiv:2409.15263](#).
10. [“Searching for magnetar binaries disrupted by core-collapse supernovae”](#)  
M. Sherman, V. Ravi, *et al.* (incl. K. Sharma), 2024, MNRAS, 534, 2, 1414-1419. [arXiv:2404.05135](#).
9. [“A Heavily Scattered Fast Radio Burst Is Viewed Through Multiple Galaxy Halos”](#)  
J. Faber, V. Ravi, *et al.* (incl. K. Sharma), under review at ApJ. [arXiv:2405.14182](#).
8. [“Polarimetry of 25 New Fast Radio Bursts Provides Insights into Their Origins”](#)  
M. Sherman, L. Connor, *et al.* (incl. K. Sharma), 2024, ApJ, 964, 131. [arXiv:2308.06813](#).
7. [“Characterizing the Ordinary Broad-line Type Ic SN 2023pel from the Energetic GRB 230812B”](#)  
G. Srinivasaragavan, V. Swain, *et al.* (incl. K. Sharma), 2024, ApJL, 960, L18. [arXiv:2310.14397](#).
6. [“Implications of Faraday Rotation Measures of Fast Radio Bursts Localized to Host Galaxies”](#)  
M. Sherman, L. Connor, *et al.* (incl. K. Sharma), 2023, ApJL, 957, L8. [arXiv:2308.06816](#).
5. [“Two Fast Radio Burst Sources in Massive Galaxy Clusters”](#)  
L. Connor, V. Ravi, *et al.* (incl. K. Sharma), 2023, ApJL, 949, L26. [arXiv:2302.14788](#).
4. [“Discovery of the Host Galaxy of FRB 20220912A”](#)  
V. Ravi, M. Catha, *et al.* (incl. K. Sharma), 2023, ApJL, 949, L3. [arXiv:2211.09049](#)
3. [“India’s First Robotic Eye for Time-domain Astrophysics: The GROWTH-India Telescope”](#)  
H. Kumar, V. Bhalerao, *et al.* (incl. K. Sharma), 2022, AJ, 164, 90, [arXiv:2206.13535](#).
2. [“Fast-transient Searches in Real Time with ZTFReST: Identification of Three Optically Discovered Gamma-Ray Burst Afterglows and New Constraints on the Kilonova Rate”](#)  
I. Andreoni, M. Coughlin, *et al.* (incl. K. Sharma), 2021, ApJ, 918, 63. [arXiv:2104.06352](#).
1. [“Time-series and Phase-curve Photometry of Active Asteroid Gault in a Quiescent State”](#)  
J. Purdum, Z. Lin, *et al.* (incl. K. Sharma), 2021 ApJL 911 L35. [arXiv:2102.13017](#)

## HONORS AND AWARDS

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- **Doug and Betty Nickerson Graduate Fellowship**, California Institute of Technology 2022-2023
- **Undergraduate Research Award: Institute Academic Prize** in recognition of exceptional research during Bachelor’s Thesis in Astronomy 2020-2022
- **ASI New Discovery Award Special Mention** from the **Astronomical Society of India (ASI)** 2020
- Achieved **99.99 Percentile** in **JEE-Advanced & Mains** national exams among 1M+ candidates 2018

## PRESS RELEASES AND SCIENCE JOURNALISM (\*INTERVIEWED)

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- California Institute of Technology press release\*, [“Fast Radio Bursts Poised to Help with Biggest Cosmic Mysteries”](#).
- Space.com feature, [“Strange ‘fast radio bursts’ across the universe could help solve major cosmic mysteries: ‘It’s only the beginning’”](#)
- IFLScience feature\*, [“Mysterious Fast Radio Bursts Could Be The Crucial Tool For Studying Undetected Matter In The Universe”](#)
- American Astronomical Society (AAS) Nova research highlights, [“Diving into the Darkest Corners of the Universe with Fast Radio Bursts”](#).
- California Institute of Technology press release\*, [“Mighty Radio Bursts Linked to Massive Galaxies”](#).
- Nature News and Views article\*, [“Mysterious radio bursts mostly come from massive galaxies”](#).
- California Institute of Technology press release, [“Missing Matter in Universe Found”](#).
- Harvard University press release, [“A New GPS for the Intergalactic Medium: Astronomers Have Found the Home Address for Universe’s ‘Missing’ Matter”](#).
- IIT Bombay press release\*, [“IITB students discover closest known asteroid to fly by Earth”](#).
- California Institute of Technology press release, [“ZTF Finds Closest Known Asteroid to Fly By Earth”](#).

- NASA JPL press release, “[Tiny Asteroid Buzzes by Earth - the Closest Flyby on Record](#)”.
- European Space Agency (ESA) article, “[2020 QG, the closest close approach](#)”.
- Forbes news article, “[Our Planet Protected Us. Earth ‘Bends’ An Asteroid Just 1,830 Miles Out, The Closest Flyby On Record](#)”.

## SCIENTIFIC WORKSHOPS AND TALKS

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### Invited Workshops

- [Baryons @ Princeton](#), Princeton University 04/15/2026
- [Exploring New Boundaries in Cosmology and Astrophysics: CMB and LSS Surveys](#), Kavli Institute for Theoretical Physics (KITP), University of California Santa Barbara (UCSB) 02/12/2026
- [FTSky: A program in the field of Fast Radio Transients](#), International Center for Theoretical Sciences (ICTS), Tata Institute of Fundamental Research (TIFR) 10/08/2025

### Invited Talks

- KIPAC Tea Talk, Stanford University 09/25/2026
- Galaxy Formation Seminar, University of California, Berkeley 09/23/2026
- Cosmology Seminar, University of California, Berkeley 09/22/2026
- Roman Collaboration Science Meeting, University of Arizona 06/12/2026
- Seminar, Harvard University 05/19/2026
- CRAFT Collaboration Science Meeting, Swinburne University of Technology 04/28/2026
- Cosmology Seminar, Princeton University 03/19/2026
- [FRB 2025 Meeting](#), McGill University 07/08/2025
- Observers Lunch, Northwestern University 05/23/2025
- Astronomy-Cosmology-Gravity (ACG) Seminar, Department of Physics, IIT Bombay 05/07/2025
- FRB Frontiers Meeting, University of Pittsburgh 03/03/2025
- Coffee Talk and Cosmology Seminar, Princeton University 02/27/2025
- Radio Lunch Seminar, California Institute of Technology 12/03/2024
- Time Domain Astronomy Meeting, Northwestern University 18/07/2023
- FRB 2023 Meeting, India 11/07/2023
- Colloquia at TIFR, National Center for Radio Astrophysics, and IIT Bombay 11/03/2023
- The Astrophysics of FRBs, Center for Computational Astrophysics, Flatiron Institute 09/11/2023
- Open Compact Binary Coalescence Call, LIGO Scientific Collaboration [LIGO-P2200227](#) 06/28/2022

### Contributed Talks

- American Astronomical Society, AAS Meeting #248, Pasadena 06/16/2026
- [New Synergies in Multi-Probe Cosmology](#), KITP, UCSB 02/06/2026
- [Cosmic Ecosystems Meeting](#), Perimeter Institute, University of Waterloo 08/01/2025
- FRB 2024 Meeting, Thailand 11/04/2024
- American Astronomical Society, AAS Meeting #243, New Orleans 01/10/2024
- Scientific Frontiers and Synergies for the DSA-2000 Radio Camera Conference, Caltech 03/20/2023
- [23rd International Conference on General Relativity and Gravitation \(GR23\)](#) 07/07/2022
- [Europlanet Science Congress 2021](#), European Planetary Science Congress 09/13/2021

### Contributed Workshops

- New opportunities in Astronomy, Owens Valley Radio Observatory (OVRO), Bishop 05/13/2025
- Multi-wavelength follow-up of FRBs in the era of routine localizations, Univ. of Toronto 04/25/2023
- Pro-Am Comet Community Workshop, Europlanet Society, Stefanik Observatory, Prague 06/10/2022
- Zwicky Transient Facility Summer School, California Institute of Technology 08/16/2021

- Radio Astronomy Winter School, NCRA, TIFR and IUCAA 12/20/2020
- Quantum Computing Workshop - an IBM Quantum Experience, IIT Bombay 07/01/2020

## RESEARCH MENTORSHIP EXPERIENCE

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- Mariana Vale Taveira (Undergraduate Student, Caltech), *Resolving the star-formation history, metallicity, and interstellar media of FRB hosts with KCWI integral field spectroscopy* 2025-Present
- Harsh Choudhary (Graduate Student, IITB), *Improving asteroid astrometry with Astreaks* 2022-2023
- Vishwajeet Swain (Graduate Student, IITB), *GROWTH-India non-sidereal observations* 2021-2022

## LEADERSHIP & SERVICE

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- **Peer Reviewer**, reviewed **10+** manuscripts for ApJ, ApJL, MNRAS, and JCAP, providing critical scientific feedback to support rigorous peer review and publication 2024-Present
- **Graduate Student Interviewer**, Caltech Astronomy Graduate Program Admissions, interviewed prospective graduate students and contributed to holistic evaluation and admissions decisions 2026
- **Graduate Student Representative**, coordinated with faculty and led the execution of department-wide Caltech Astronomy colloquia, facilitating departmental scientific exchange 2024-2026
- **Lead Organizer**, Caltech Astronomy Prospective Graduate Student Visit, coordinated departmental programming and logistics for prospective graduate students visit 2024
- **Graduate Student Mentor**, supported incoming graduate students in navigating the academic, professional, and community aspects of graduate life at Caltech 2024

## TEACHING & OUTREACH

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- **Caltech Astronomy Outreach Volunteer**, Q&A panelist for star-gazing lecture 2025
  - Engaged with the community and **communicated astronomy research** in an accessible way
  - Contributed to public science outreach through **interactive discussions and Q&A** on radio astronomy, the DSA-2000 survey, and the science of the night sky
- **Lecturer**, *FTSky: A program in the field of Fast Radio Transients Workshop*, ICTS-TIFR 2025
  - Led an intensive **hands-on training session** on optical studies of FRB host galaxies, covering archival data mining, data reduction, galaxy morphology, and SED analysis
  - Designed and **delivered lectures/tutorials** on probabilistic host-galaxy association, optical spectroscopy, and SED modeling, **guiding 100+ students** through astronomical analysis workflows
- **Teaching Assistant of Graduate Level Courses**

*Conducted tutorials, graded assignments, mentored students and supported smooth course delivery*

  - Ay124: *Structure and Evolution of Galaxies*, Prof. Phil Hopkins, Caltech Winter 2024
  - Ay121: *Radiative Processes*, Prof. Sterl Phinney, Caltech Fall 2023
  - PH556: *Astrophysics*, Prof. Varun Bhalerao, IIT Bombay Spring 2021
- **Maunakea Astroday Public Outreach Program Volunteer** 2024
 

*Astronomer Community Engagement (ACE) Program, Caltech Optical Observatories*

  - Introduced **elementary school students** to the fundamentals of UV, IR, and optical astronomy
  - Demonstrated the ability of IR in **probing dust obscured** content of star-forming galaxies
- **Lecturer**, *Scientific Computation & Mathematical Modeling Course*, IIT Bombay Summer 2020
  - Provided **hands-on Python training** to **250+ students** on solving scientific problems
  - Designed **course material** on numerical methods and algorithms for differential equations
- **Project Moderator**, *Summer of Science reading projects*, IIT Bombay 2020-2021
  - Advised a group of **6 students**, guiding them for a three month long **reading project**
  - **Fostered active discussions** within the group by sharing resources on observational astronomy
- **Amatuer Astronomy Club Manager**, *Krittika, The Astronomy Club of IIT Bombay* 2019-2021
  - Managed **10 conveners**, tending to a community of  $\sim 100$  at IITB and  $\sim 5,500$  enthusiasts online
  - Supervised **6 projects**, 2 week-long tutorials on **numerical computing** and 3-day workshops

- Ideated **Team ANYmations** to generate scriptable scientific animations for publicity campaigns
- Organized **institute-wide events** such as lectures, discussions, documentary screenings, and competitions such as **Astromania, Scientific Computation/Observation Planning** Championships
- Sensitized enthusiasts to **astrophotography** by conducting night sky **observation sessions** and familiarized beginners with basic concepts of telescope handling and observation planning

## TECHNICAL SKILLS

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Relevant first-author publications demonstrating astrophysics expertise denoted with [.]

**Astrophysics expertise** Hydrodynamical simulations analysis [6], multi-probe cosmological analyses [10, 11], Fisher forecasting [7], Bayesian modeling/statistical inference [9], ML/time-series analysis [3], astronomical software development [2], galaxy SED modeling [4, 1]

**Languages** Advanced proficiency in Python, with experience in Git, C++, SQL and MATLAB

**ML/AI Stack** PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, OpenCV, SLURM

## PROFESSIONAL REFERENCES

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|                        |                                                  |                                                                              |
|------------------------|--------------------------------------------------|------------------------------------------------------------------------------|
| Dr. Vikram Ravi        | Professor of Astronomy, Caltech                  | <a href="mailto:vikram@astro.caltech.edu">vikram@astro.caltech.edu</a>       |
| Dr. Elisabeth Krause   | Professor of Astronomy, University of Arizona    | <a href="mailto:krausee@arizona.edu">krausee@arizona.edu</a>                 |
| Dr. Jason X. Prochaska | Professor, University of California Santa Cruz   | <a href="mailto:xavier@ucolick.org">xavier@ucolick.org</a>                   |
| Dr. Simone Ferraro     | Scientist, Lawrence Berkeley National Laboratory | <a href="mailto:sferraro@lbl.gov">sferraro@lbl.gov</a>                       |
| Dr. Liam Connor        | Assistant Professor of Astronomy, Harvard        | <a href="mailto:liam.connor@cfa.harvard.edu">liam.connor@cfa.harvard.edu</a> |
| Dr. Boryana Hadzhiyska | Assistant Professor, University of Cambridge     | <a href="mailto:bth26@cam.ac.uk">bth26@cam.ac.uk</a>                         |
| Dr. Varun Bhalerao     | Associate Professor, IIT Bombay                  | <a href="mailto:varunb@iitb.ac.in">varunb@iitb.ac.in</a>                     |
| Dr. Archana Pai        | Associate Professor, IIT Bombay                  | <a href="mailto:archanap@iitb.ac.in">archanap@iitb.ac.in</a>                 |